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NFTX – Good Idea, Bad Design

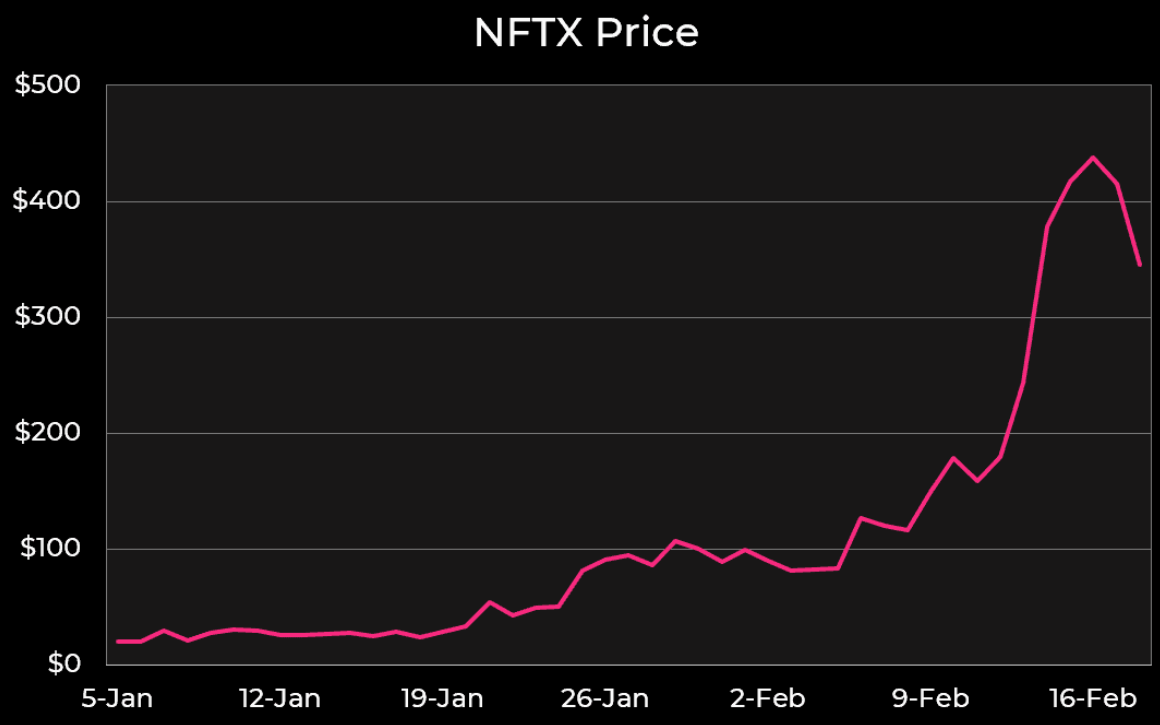
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Guest Analyst Coalie Bee

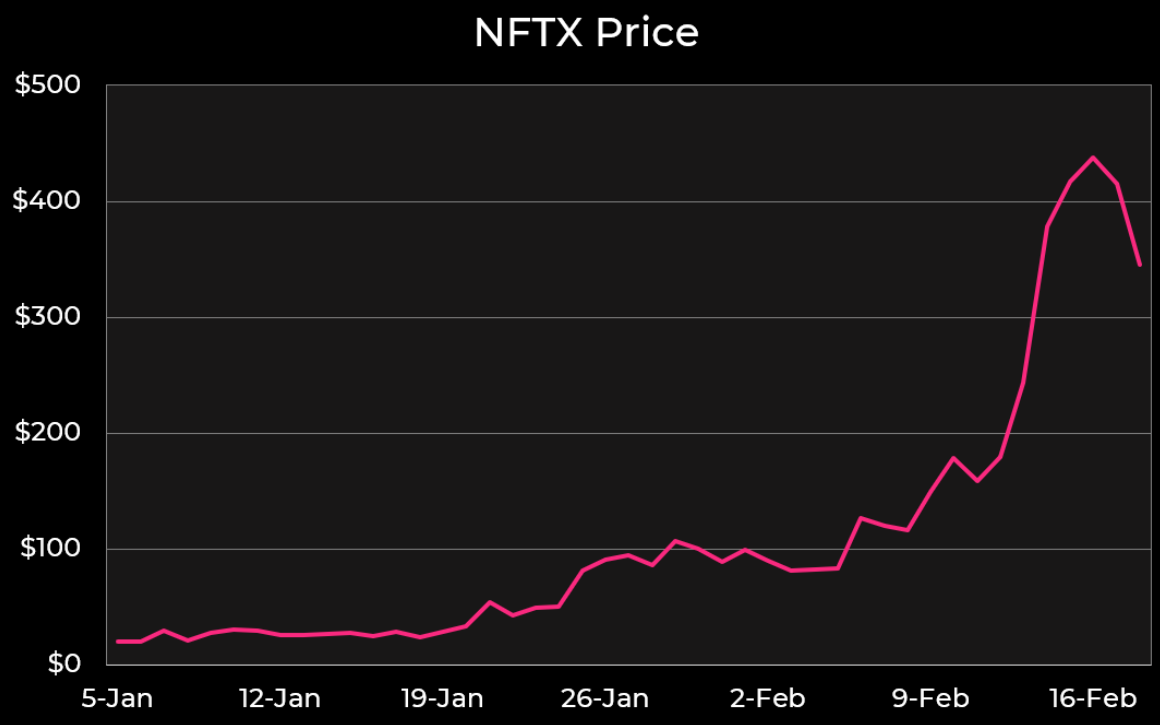
Axie Infinity | AXS | Mystic Axie | NFT / Gaming | Nftx | Yield Guild

Disclosure: Delphi Ventures and/or members of our team have invested in Axie Infinity and Yield Guild. This statement is intended to disclose any perceived conflict of interest and should not be misconstrued as a recommendation to purchase any tokens. This is not investment advice.



It's no secret that Delphi has been bullish on the NFT space for some time, having worked closely with Axie Infinity to consult on, and invest in, the game's governance token. Late last year, we doubled down on our conviction by purchasing a portfolio of highly sought after Mystic Axies. More recently, we invested in Yield Guild which we believe to be a promising project operating across this ecosystem. Other engagements are also in the works. That said, we still believe the NFT market to be very early and are concerned that investors in the ecosystem may be getting carried away with the current hype cycle.

As mentioned in our note from earlier this week, it seems that things are getting very heated in the NFT sector. People are hungry for exposure, but it's not clear that the space is mature enough yet to absorb all of the capital being thrown at it. The use cases are beginning to be well-defined, but the paths to their fulfillment less so. Delphi's investments in the space are limited to a handful of concentrated bets in standout projects that have stood up to rigorous scrutiny from our team. Much in the same way not all projects survived crypto's last big hype cycle, we believe many of the NFT-related projects currently garnering attention have fundamental flaws that could prove difficult to resolve, albeit not impossible. One such example is NFTX, which we shall examine below.



Firstly, let's run through how NFTX works. It's an NFT indexing platform for turning baskets of NFTs into ERC20 tokens which can then be traded with all the benefits of fungibility. There are 2 types of funds, D1 and D2. D1 funds are made up of the NFTs themselves and D1 tokens represent 1:1 backing of NFTs in the fund. For example, if you put a single mystic Axie in the MYSTIC-AXIE-1 pool, you will be given a MYSTIC-AXIE-1 D1 fund token. At any time in future, you or someone else may trade this token into the MYSTIC-AXIE-1 pool to retrieve a **random** single mystic axie from the pool. In this way D1 tokens represent ownership of NFTs in the pool at a 1:1 ratio. D1 tokens then act as building blocks for D2 Funds, where many D1 tokens may be pooled together to create a more diversified index of assets.

The first fundamental issue arises when we look at the mechanics of D1 tokens and how they value assets within D1 funds. **In the eyes of a D1 token, every asset in its D1 fund has exactly the same value. This is suboptimal in our opinion.** The problem stems from the arbitrage opportunity that emerges when assets in the pool have varying values. Either from the start, or ones which emerge over time as asset values change. For example, the fund mentioned above, MYSTIC-AXIE-1 which is for single mystic Axies, involves assets who range in value from 5 eth all the way up to 100 eth for the most sought after single mystics, like those who are exceptional battlers or have extremely low IDs.

In this case, there is never an incentive for anyone to add a mystic with above average value to the pool and if one is added, **there is a large incentive for someone to come in, simply purchase a floor axie for 5 eth and repeatedly trade it in and out for D1 tokens until they randomly hit the high valued Axie.** Not only are these pools exclusively suited to floor assets, but overtime as some of the assets actually move up from floor prices, they are traded away for less valuable assets, leaving fund owners holding the bags of the cheapest assets in each category and nothing to really show for it in terms of yield. You would literally be better off buying almost any single mystic axie in the long run, than buying exposure to an NFTX MYSTIC-AXIE-1 pool, because **the pool's average value will always trend toward the floor.** MUSE is another NFT index project that works in a similar way, but each NFT is represented by 100 tokens and you can actually choose the NFT you wish to remove from the pool, making these problems even worse.

This creates a situation where D1 funds only work for incredibly narrow groups of assets, to the point it basically defeats the purpose of an index in the first place. D2 Funds then seem to be an attempt to bandage this critical issue, by allowing you to create very narrow funds on the D1 layer and create diversification on the D2 layer. However, D2 tokens fail to efficiently create diversification because assets will constantly need to be recategorized into different D1 funds as asset values diverge and parameters need to be changed anyway, destroying any current plan to create narrow tranches on the D1 Layer and simply organize them on the D2 layer.

As a simple example, let's take a pool with just 2 types of assets in it. If the price of both assets started at $100, the pool value is $200, and there are two D1 tokens. Now, let's say the value of asset A drops to $50, whilst the price of asset B increases to $150. It now makes sense for people to buy more of asset A for $50 (very similar NFT which fits with the same meta data characteristics) and trade it into the pool. This gives you a 33% chance at receiving an asset worth $150, and 66% chance at getting a $50 asset back. If you fail, you may repeat the process until you get the $150 asset. Now, the pool has 2 assets in it, is worth $100, and our trader has walked away with 100 dollars, leaving the other 2 D1 token holders with the bags and a bunch of floor assets where they only got effective exposure to the asset that was losing value, and no exposure to the asset gaining value.

This problem continues as you scale up, just requiring more trades and swaps to get the asset you want, and continues to exacerbate as prices continue to diverge. There is no way to know when you originally set the parameters for a given D1 fund that those assets stay like-kind in the eyes of the market, meaning parameters must be changed and adjusted. This creates an unacceptable amount of complexity. This model fundamentally fails to capture value for both token holders and fund participants in the long run, with the majority of gains being arbitraged away by traders.

Other issues for NFTX include a poor token distribution with a large portion of tokens going to a handful of NFT/Cryptopunk whales and 10% going to the founder, with nothing set aside for a treasury to fund other team members and future development. Poor initial distribution combined with a lack of future treasury funding may prove to be major issues for NFTX long term.

The Solution

There are solutions on the horizon, however, and the NFT index use case will no doubt be a massive opportunity. We believe it's just too early for this key infrastructure to emerge as we need to see development of NFT valuation games such as Upshot. These are systems of incentives for people to provide accurate data and guesses for what they believe the value of NFTs are on an ongoing basis, in order to give the market and smart contracts a point of reference to make decisions about NFT values. Wisdom of the crowd mixed with some clever AI and big data algorithms will likely be most effective here. Without this core piece of infrastructure, many critical NFT use cases are doomed.

It's quite likely that a valuation game is so crucial to the NFT index use case that the valuation game itself and its participants should be integrated at the core of the project and token economics, so the people ultimately being paid fees from the indices, are the people contributing to the valuation of the assets in the indices. The valuation of NFTs in the indices is such a core part of the use case that there is little need for 2 separate value accrual tokens, when they can essentially be made one, and right now those collecting fees from the NFT indices don't really provide ongoing value. This also creates a situation with a kind of internal oracle being generated where other projects can plug into the

valuations being indicated by these indices to derive an idea of NFT value. Combining this with strong algorithms, metadata capabilities and yield farming for a potentially very accurate NFT valuation game with strong incentive alignment and network effects.

There are likely several effective solutions to the valuation game problem and we are excited to follow this segment as it emerges over the coming years and believe it may become a multi trillion dollar category over the coming decades as markets like real estate get the NFT treatment. However, we believe the existing projects are too early and primitive to truly manifest this use case, and incentives must be figured out, along with algorithms to maximize the effectiveness and accuracy of data collected.

Could existing platforms successfully pivot to these new models in the future? Potentially, but it will be very difficult for them. Everything would need to fundamentally change about the project structures right now, from smart contracts to token economics. There is a kind of technical debt here and combined with the burden of poor token distributions might leave the early NFT platforms unable to shift effectively. The only thing they will really have going for them is some established network effects, which are not all that strong right now anyway. It's going to be an uphill battle for the projects who rushed to launch in our opinion.